

Directions for using the Functional Performance Test Module (FPTM) and BACnet Data Source (BDS) software (DRAFT)

The BDS is a software tool developed by NIST that can be used to discover BACnet points on physical controllers. This point information is provided to the FPTM as part of the workflow for the point mapping process. This process connects points on a controller to the point names that are referenced in the test scripts.

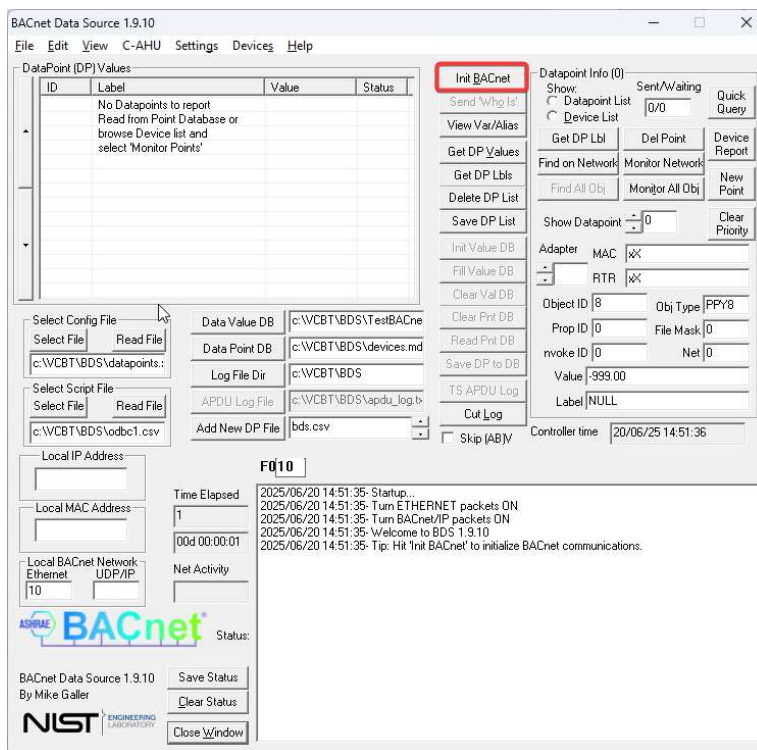
The FPTM is a tool that will read the XML test scripts and execute scripted tests of controller programming over BACnet.

Installation:

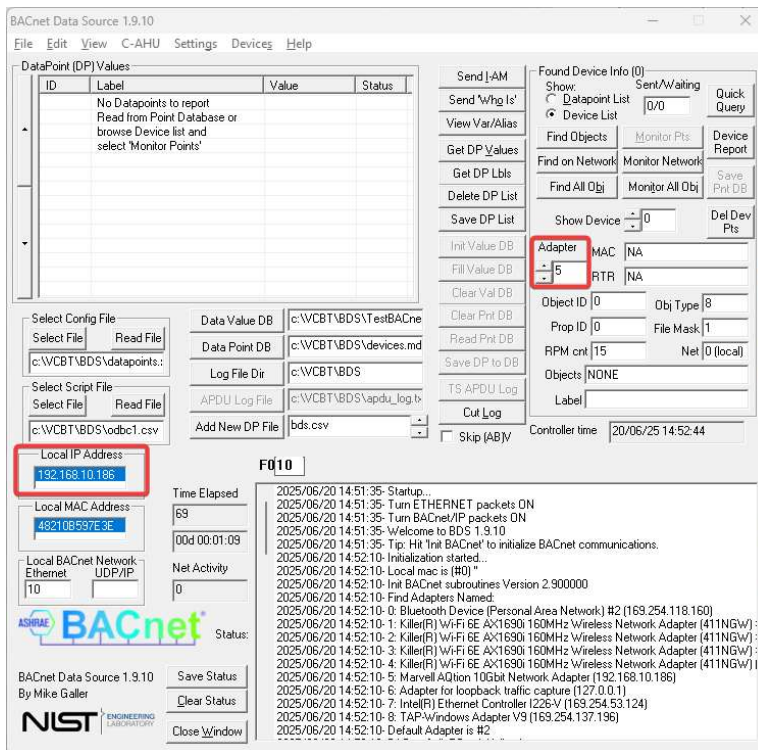
1. Download the “Functional Performance Test Module.exe” and “BDS.exe” files to a folder on your local hard drive. Put each executable file in a separate directory. Copy the “BACdll2012.dll” file to both directories.
2. Download and install the latest version of Npcap: <https://npcap.com/dist/>

Point Mapping:

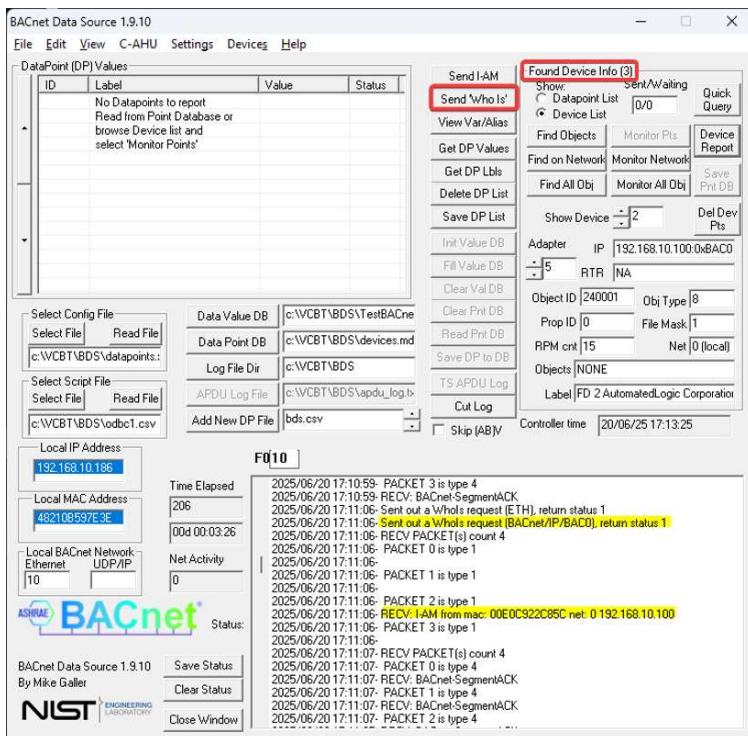
3. Point map file: The FPTM can create a point mapping file using a file generated by the BDS. To generate this file, connect computer to a controller that is hosting the program to be tested. Open the BDS and:
 - a. Click on the “Init BACnet” button.



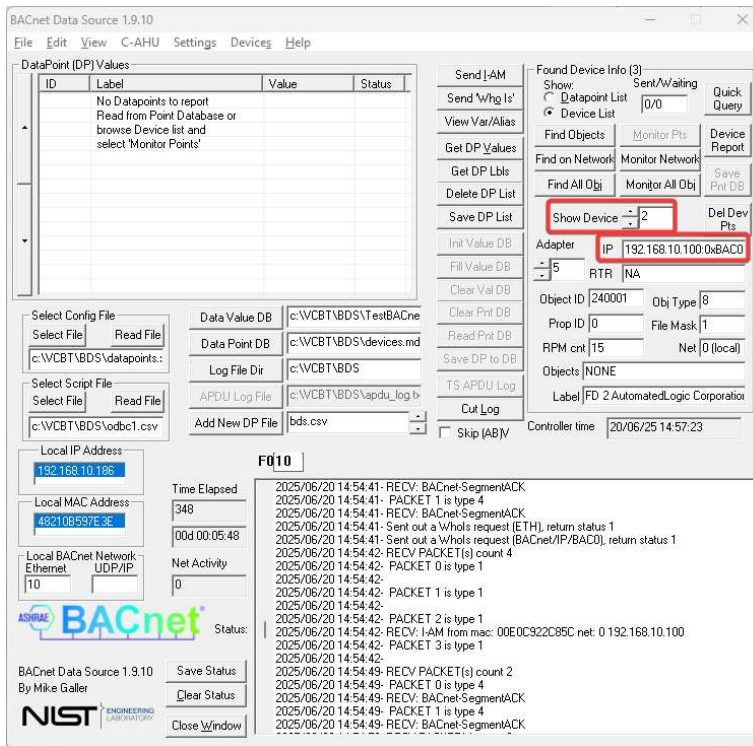
- b. Check that the IP address of the computer running the BDS is shown in the “Local IP Address” box on the left side of the BDS. If not, find the “Adapter” box on the right side and click the up/down arrows until the correct IP address is shown.



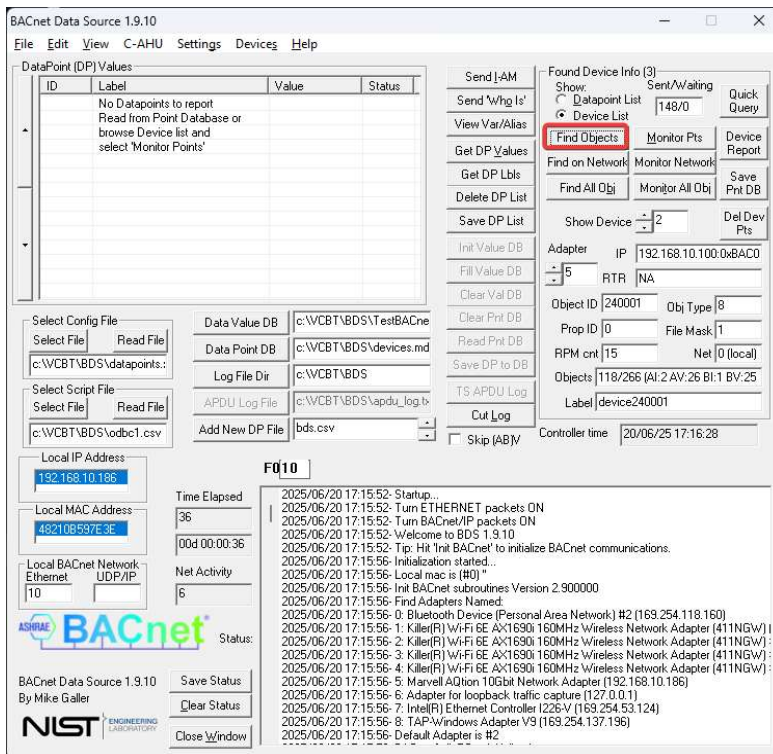
- c. Click on the “Send ‘Who is’” button. The BDS should receive an “I-Am” message from the controller, which will be displayed in the status text box. The count on the “Found Device Info (##)” display should also increase.



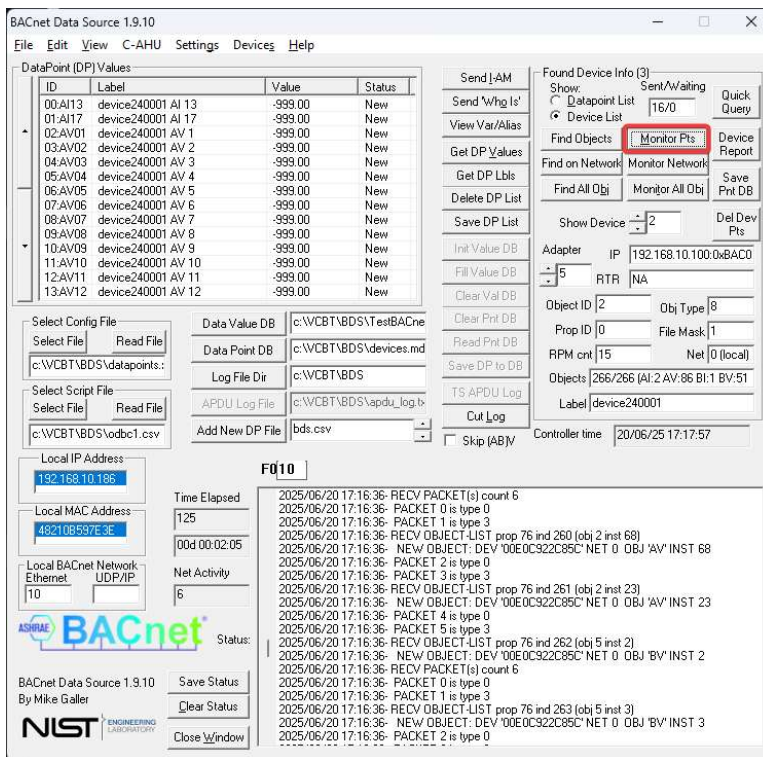
- d. If more than one controller is on the network use the up/down arrows next to the “Show Device” label until the information for the test controller is displayed in the text boxes below the label.



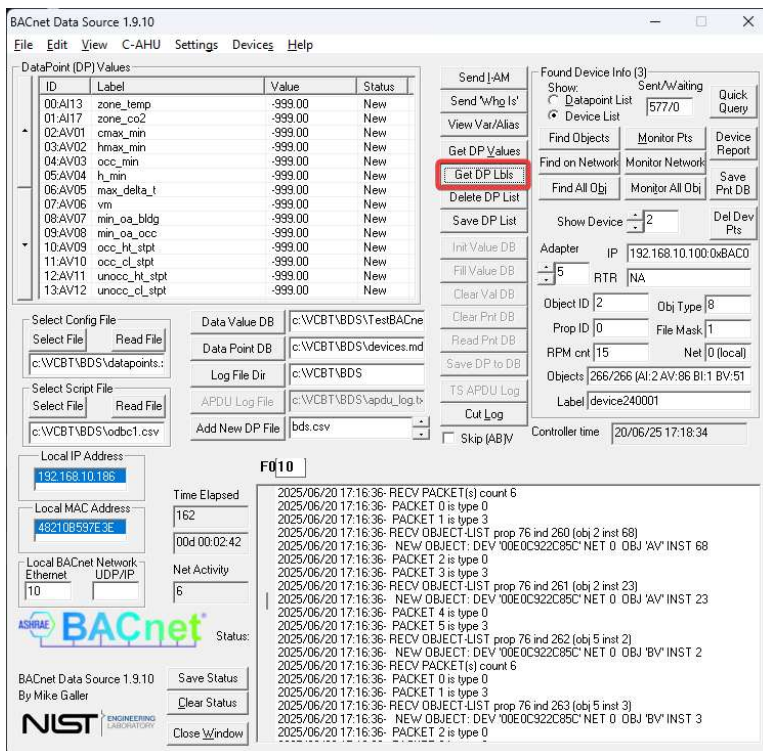
- e. Click on the “Find Objects” button to query the controller points. Wait until the counter in the “Sent/Waiting” box is 0 before going to the next step.



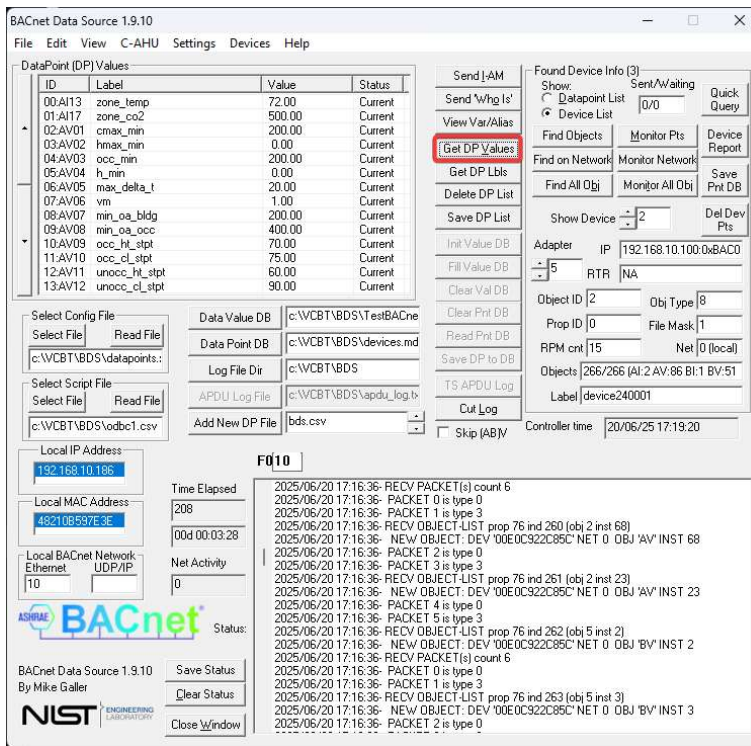
- f. Click on the “Monitor Pts” button. This will add all points of monitored object types (AI, AO, AV, BI, BO, BV, MSI, MSO, MSV) to the DataPoint list. Wait until the counter in the “Sent/Waiting” box is 0 before going to the next step.



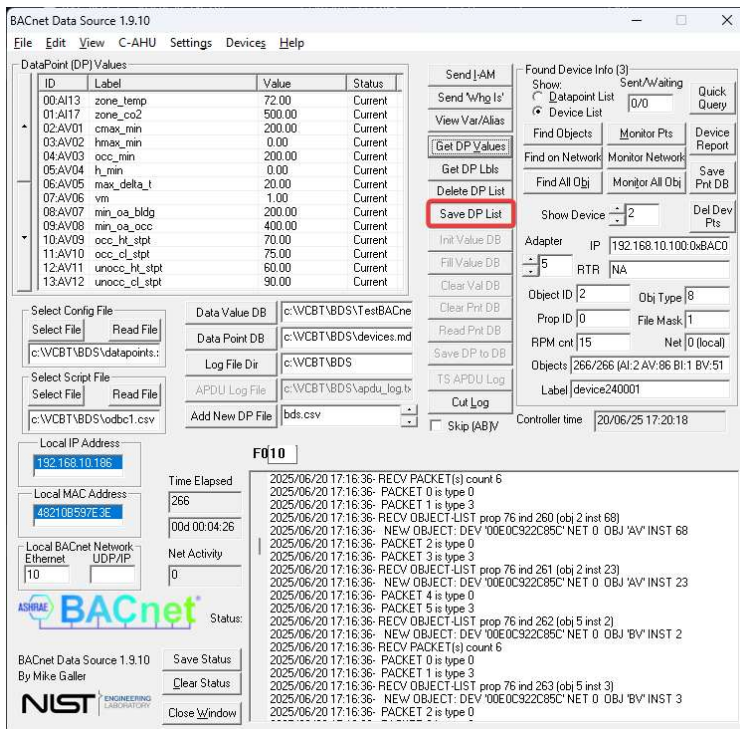
- g. Click on the “Get DP Lbls” button. This will retrieve the labels (BACnet point names) and description property data from the objects in the DataPoint list. Wait until the counter in the “Sent/Waiting” box is 0 before going to the next step.



- h. Optionally click on the “Get DP Values” button to retrieve the present-value property from the objects in the DataPoint list. The BDS can also optionally save time series values to a CSV file by selecting “Settings->BACnet on Timer”.



- i. Click on “Save DP List” and follow the prompts to save the DataPoint list to a BDS config file in XML format. This file can be used to provide point mapping information by dropping it onto the FPTM.



- j. Drag and drop the BDS config file onto the FPTM. Check the status window for errors. If there are no errors then select “File->Save Event File” to save the FPTM point mapping file. Alternatively, use the sample file “map file for FPTM.xml” as the FPTM point mapping file.
- k. Manual step: open the FPTM point mapping file in a text editor or an XML editor. There are a series of “point” elements in the file, each containing “label” and “description” attributes, each with a value (i.e., label=“zone_temp”) describing the intent of that element. Manually add a

new “alias” attribute with the value selected from the “Control Point Name” column in the script file, as shown in bold below. Note that these attributes are case sensitive. Depending on how the controller is configured it is probable that there will be “point” elements that don’t have a corresponding item in the script file, but each item in the “Control Point Name” column must be added to the FPTM point mapping file. Reference the sample file if there are any items that are difficult to match. Save the FPTM point mapping file.

```
<point label="zone_temp" id="13" description="Zone Temperature" type="AI" alias="ZoneTemp">
<point label="da_temp" id="14" description="Discharge Air Temperature" type="AI" alias="ZoneDAT">
<point label="zone_co2" id="17" description="Zone CO2 Level" type="AI" alias="ZoneCO2">
<point label="custom" id="5" description="not matched" type="MSV">
```

- l. The FPTM will read this modified FPTM point mapping file to map read/write commands to the associated BACnet points on the controller. The FPTM will use the value of the point ID (“id”) and point type (“type”) attributes as the BACnet object instance and type. The FPTM will compare the values of the “label”, “description,” and “alias” attributes against the labels found when a script file is entered. Do not edit the FPTM point mapping file in Excel as it does not render the XML file correctly.

Executing tests with FPTM:

4. Working with test script files: The FPTM reads test script files saved in an XML format. To save a new test script file based on an Excel version: Select File->Save As, then change the type to “XML Spreadsheet 2003 (*.xml)”.
5. Running the FPTM: Open the FPTM and:
 - a. Drag and drop the point mapping file on the FPTM. Check the status window for error reports.
 - b. Drag and drop the test script file on the FPTM. Check the status window for error reports.
 - c. Use the arrows next to “Select Event Block” to select the desired event block.
 - d. Click on the “Reset Times” button, and then the “Start” button.
 - e. The FPTM will start the selected set of tests. It will move to the next set when finished. It will not loop back to lower number tests.
 - f. The FPTM will display progress and events in the status window. While running a test script the event configuration inputs are disabled and can not be modified.
 - g. When the run is complete the event configuration inputs will be enabled and the “Stop” button will change back to “Start”.
 - h. Note that the contents of the status window are saved to a text file named “FPTM-STATUS-log-(date)-(time).txt where (date) is the date in MMDDYY format and (time) is the local time in HHMMSS format. This file is saved automatically when the run is complete.